

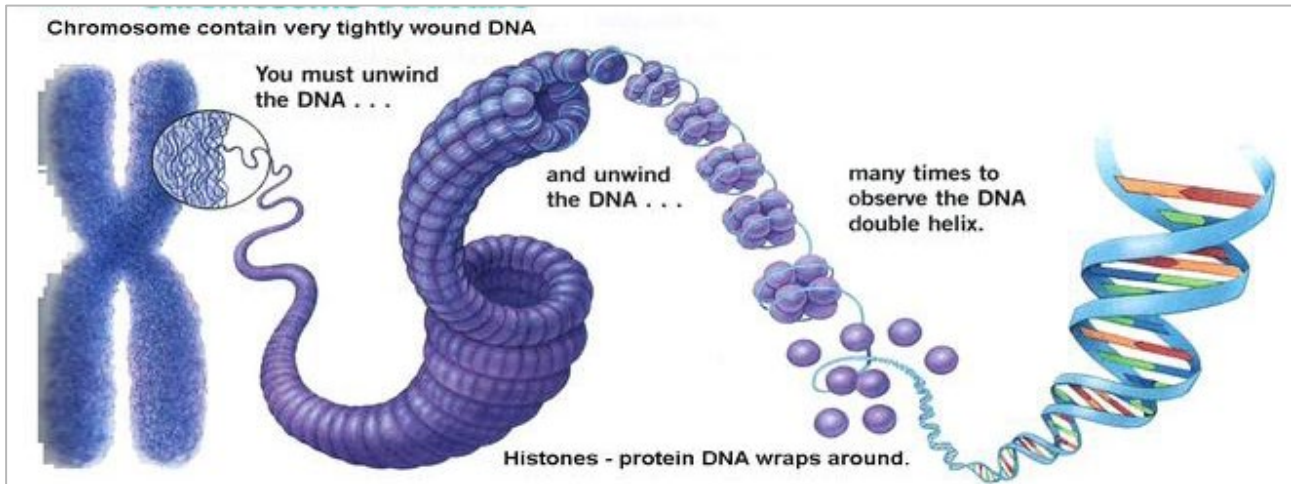
Name: _____

UNIT 3A • CELL DIVISION

Topic 1: Mitosis

Chromosomes & Cell Division

DNA in the nucleus of each cell is tightly packaged into structures called chromosomes:



- *Explain the difference between chromosomes and chromatin. When would each be visible within a given cell?*

Chromosome Number

- Chromosome number varies from species to species.
- In humans, somatic cells contain _____ chromosomes (they are diploid, or _____), while gametes contain _____ chromosomes (they are haploid, or _____).
- Species which contain more than two copies of each chromosome are _____.

Name: _____

Define each of the following terms in the context of the human genetic material:

- Somatic cells
- Gametes
- Homologous chromosomes
- Autosomes
- Sex chromosomes
- Alleles
- Diploid
- Haploid

Name: _____

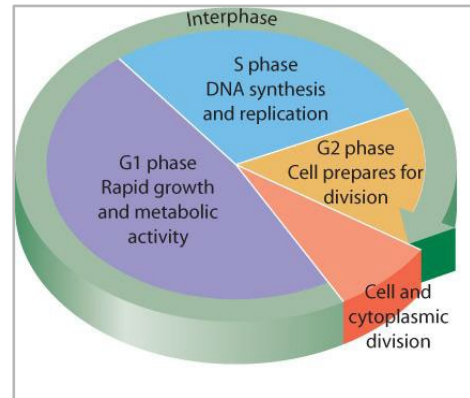
The Cell Cycle

The goal of the cell cycle is to produce daughter cells with the same DNA as the parent cell. Each cell goes through three distinct phases before it actually divides:

1)

2)

3)



Collectively, these three phases are referred to as _____.

Interphase is followed by _____, or

“_____”.

M-phase accounts for the actual division of genetic information (_____) and splitting of the cytoplasm itself (_____).

Mitosis

Mitosis is performed by all somatic cells. It is a form of cell division that produces daughter cells that are identical to parent cells. List and explain the three general purposes of mitosis:

•

•

•

Name: _____

After DNA replication (interphase), cells must go through four stages of mitosis before cytokinesis can occur...

1. PROPHASE

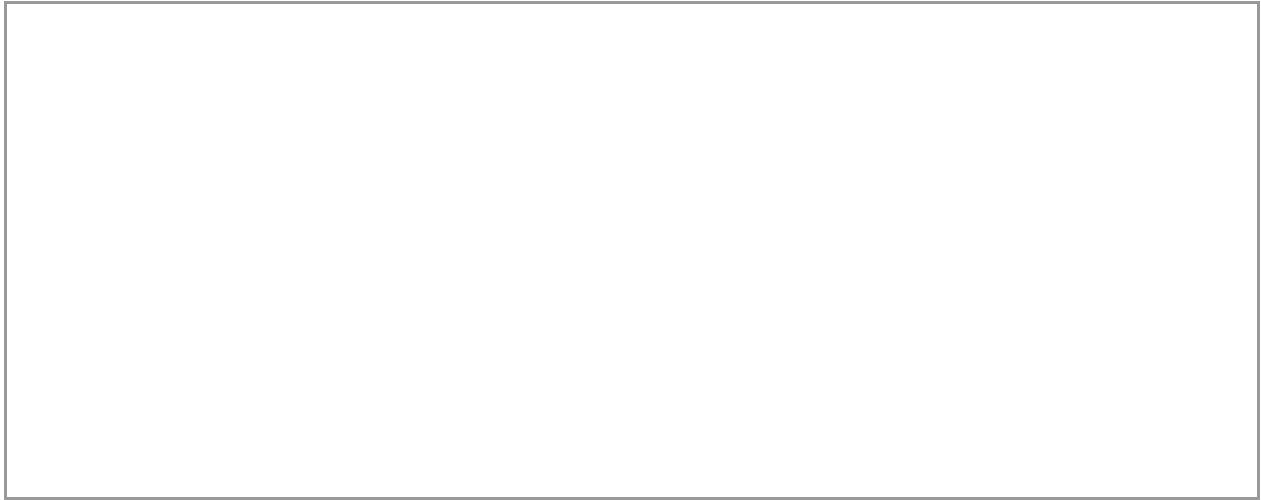
2. METAPHASE

3. ANAPHASE

4. TELOPHASE

Name: _____

- *Explain the difference between homologous chromosomes and sister chromatids. Use a diagram to illustrate your answer.*



- *How does mitosis in animal cells differ from mitosis in plant cells?*

Mitosis in Animal Cells	Mitosis in Plant Cells

Applications of the Cell Cycle

Thanks to an in-depth understanding of the cell cycle, we have been able to develop a number of useful technologies, including:

- 1)
- 2)
- 3)

Name: _____

4)

1) CANCER TREATMENTS

- *Explain how you know if a sample of cells is cancerous.*

- *Distinguish between mutagens and carcinogens.*

2) CLONING

- *Explain the difference between monozygotic and dizygotic twins.*

- *Use a flowchart to explain the process involved in artificially cloning an organism:*

Name: _____

3) STEM CELL RESEARCH

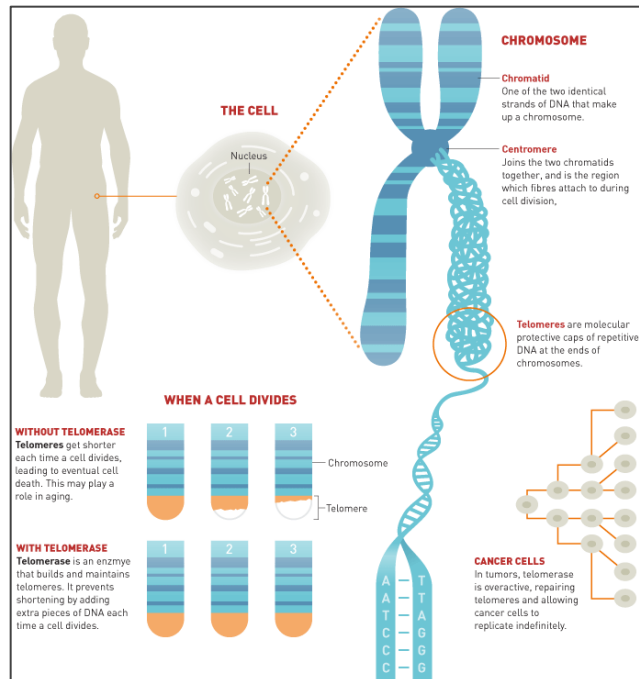
- *What is a stem cell? Outline some of the uses of stem cell research.*

- *Distinguish between totipotent and pluripotent stem cells.*

4) TREATMENTS FOR AGING?

- *How can telomeres be used to determine a cell's age?*

Name: _____



Name: _____

UNIT 3A: CELL DIVISION

Topic 2: Meiosis

Read pages 57286 in your textbook before completing the following notes.

While _____ cells contain a complete set of chromosomes, _____ possess just half the genetic information of parent cells. This is accomplished through the process of _____.

Meiosis

The outcome of meiosis is different than that of mitosis in three ways:

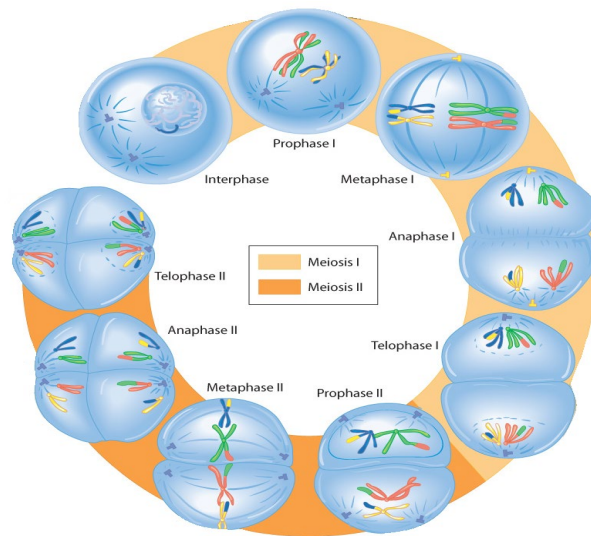
1)

2)

3)

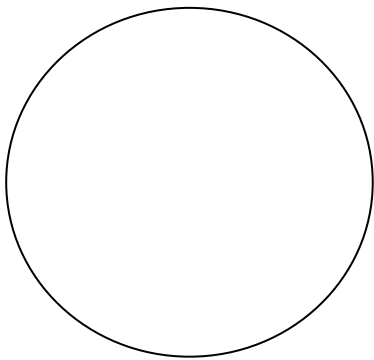
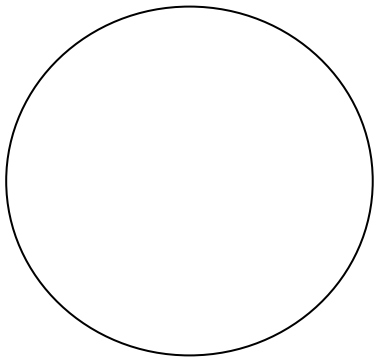
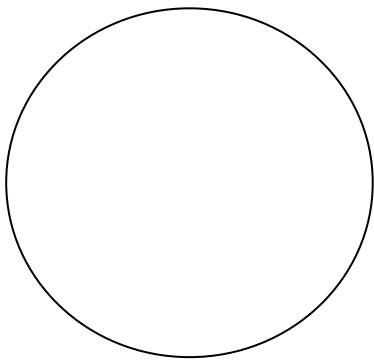
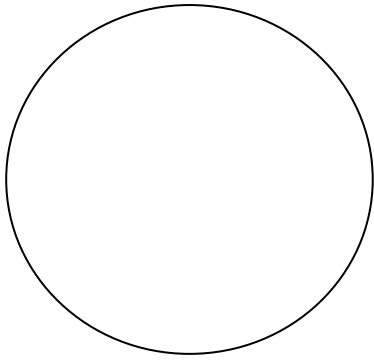
Although meiosis involves the same stages of cell division as mitosis, each stage occurs twice...

Name: _____



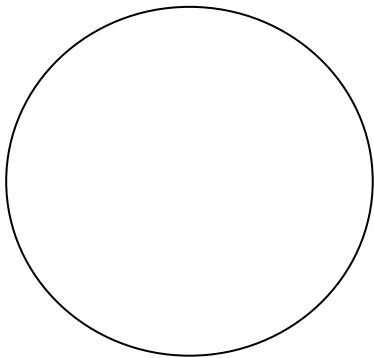
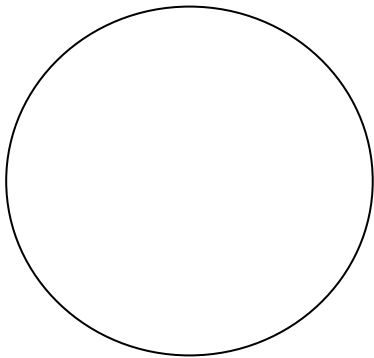
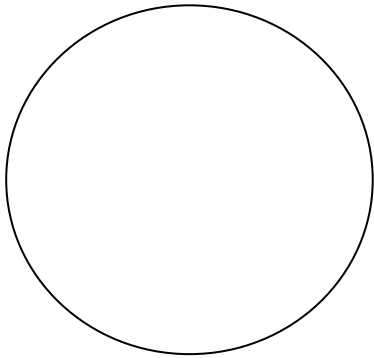
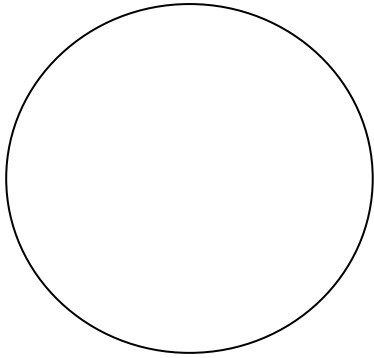
Meiosis I

Name: _____



Meiosis II

Name: _____



Gametogenesis

Name: _____

- *How does spermatogenesis differ from oogenesis? How are these two processes similar?*

- *Why does female gametogenesis involve unequal division of the cytoplasm?*

Nondisjunction

- *Explain how nondisjunction can result in monosomy of a particular chromosome.*

- *Explain how nondisjunction can result in trisomy of a particular chromosome.*

Name: _____

UNIT 3A: CELL DIVISION

Topic 3: Alternation of Generations

Reproductive Strategies

The human life cycle relies on regular patterns of meiosis and mitosis. However, other organisms rely on alternative reproductive strategies, including the following forms of asexual reproduction:

Type of Reproduction	How it works	Organisms that use it
Binary fission		
Bacterial conjugation		
Budding		
Vegetative reproduction		
Parthenogenesis		
Spores		

➤ *Outline some advantages and disadvantages of asexual reproduction.*

Name: _____

Alternation of Generations

Some organisms have evolved to utilize both sexual and asexual reproduction at different points in their life cycle...

General model for alternation of generations:

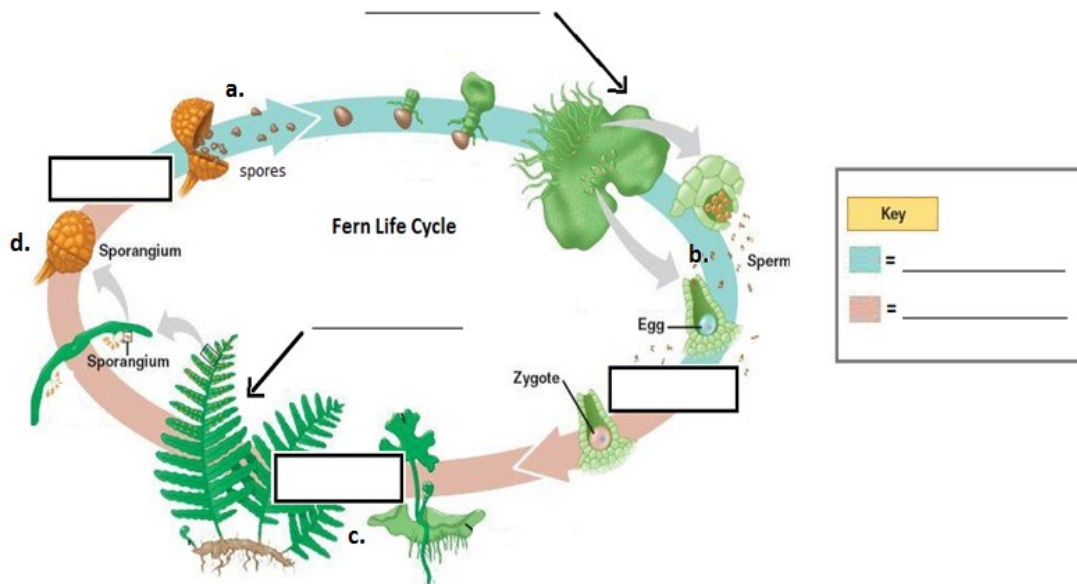
➤ *Distinguish between the gametophyte and the sporophyte in the model above.*

Name: _____

PRACTICE

Use the following terms to label the diagrams below...

Meiosis	Gametophyte	Haploid generation (n)	Fertilization
Mitosis	Sporophyte	Diploid generation (2n)	



Name: _____

